

Tourism Experience Analytics

Executive Summary

This project analyzes tourism transaction data covering 52,930 visits by 33,530 users across 9 attraction types and multiple regions worldwide. Three machine learning components were built: a regression model to predict attraction ratings, a classification model to predict visit mode (Business, Family, Couples, Friends, etc.), and a hybrid recommendation system combining collaborative and content-based filtering. All three are deployed in an interactive Streamlit application allowing users to input their profile and receive predictions and personalized attraction recommendations.

Key finding: attraction ratings and visit mode are only weakly explained by geography and timing alone (regression $R^2 \approx 0.10$, classification accuracy $\approx 49\%$), indicating that individual user experience and preference, not demographics, are the dominant drivers of satisfaction. This is itself an actionable insight: personalization strategies should weight individual history over broad demographic segmentation.

1. Data Preparation

Nine source tables were combined: Transaction, User, City, Country, Region, Continent, Attraction Type, Visit Mode, and Item (attraction) data.

Cleaning steps performed

- Stripped and standardized text fields (trimmed whitespace, title-cased categorical values) across all 9 tables.
- Dropped transaction rows with missing or invalid Rating values (kept only ratings in the valid 1-5 range).
- Dropped transaction rows with missing VisitMode, since it is a required classification target.
- Removed exact duplicate transaction records.
- Standardized VisitYear and VisitMonth as consistent integer fields for time-based analysis.
- Left-joined User to City, Country, Region and Continent, joined Item (attraction) to Attraction Type, and joined Transaction to the Visit Mode lookup, into a single master dataset.
- Filled remaining categorical gaps (8 rows missing city or country information after merging) with 'Unknown' rather than dropping, a negligible fraction of the 52,930-row dataset.

Result

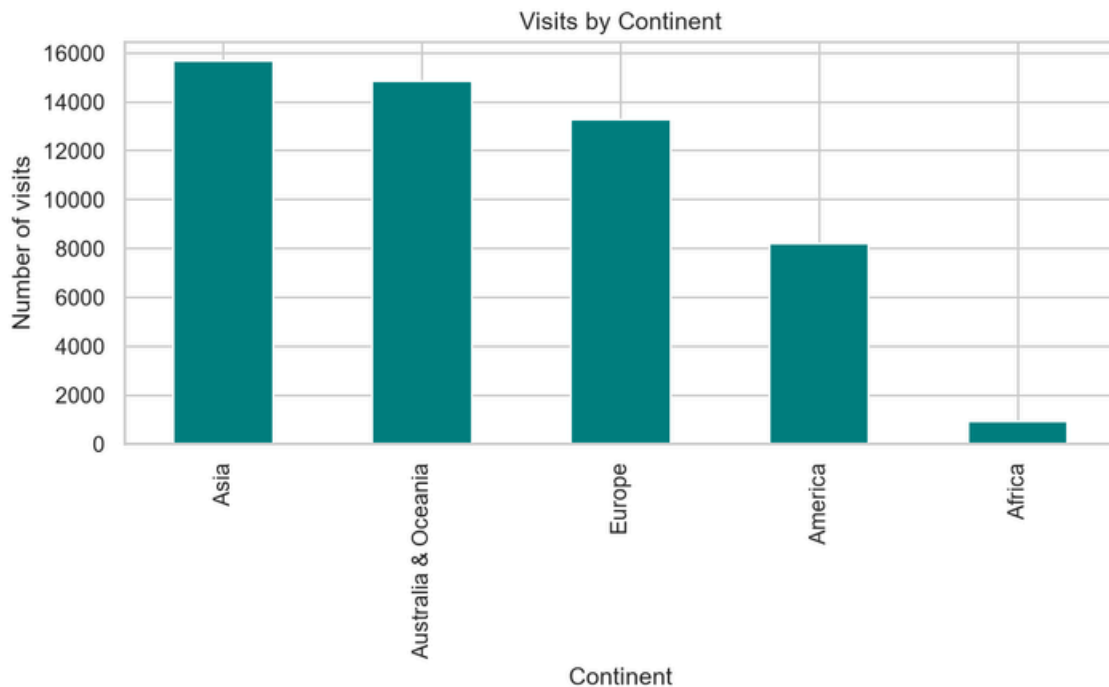
<i>Metric</i>	<i>Value</i>
Raw transaction rows	52,930
Rows after cleaning	52,930
Unique users	33,530
Final merged dataset	52,930 rows x 23 columns
Missing values after merge	8 rows (city/country), filled as 'Unknown'

2. Exploratory Data Analysis

Eight visualizations were produced covering user distribution, attraction popularity, rating patterns, and seasonality. Key numeric findings from the attraction-type breakdown are summarized below, followed by each chart with its interpretation.

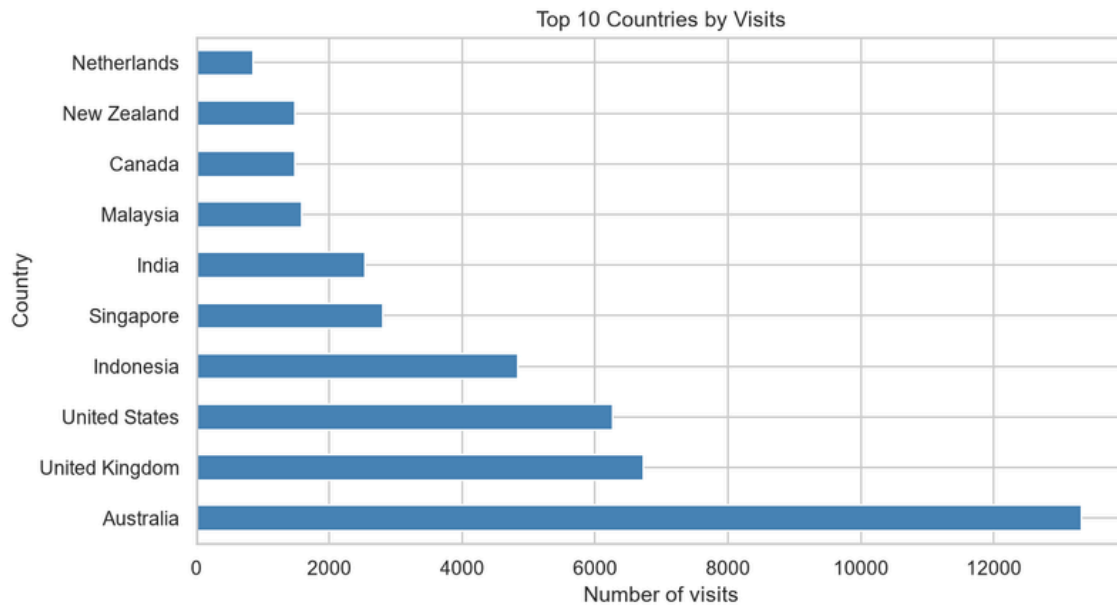
<i>Attraction Type</i>	<i>Avg Rating</i>	<i>Visit Count</i>
Water Parks	4.65	6,429
Caverns & Caves	4.50	135
National Parks	4.42	511
Nature & Wildlife Areas	4.27	13,251
History Museums	4.26	978
Religious Sites	4.21	6,711
Points of Interest & Landmarks	4.13	6,252
Beaches	3.85	10,917
Historic Sites	3.54	798

Chart 1: Visits by Continent



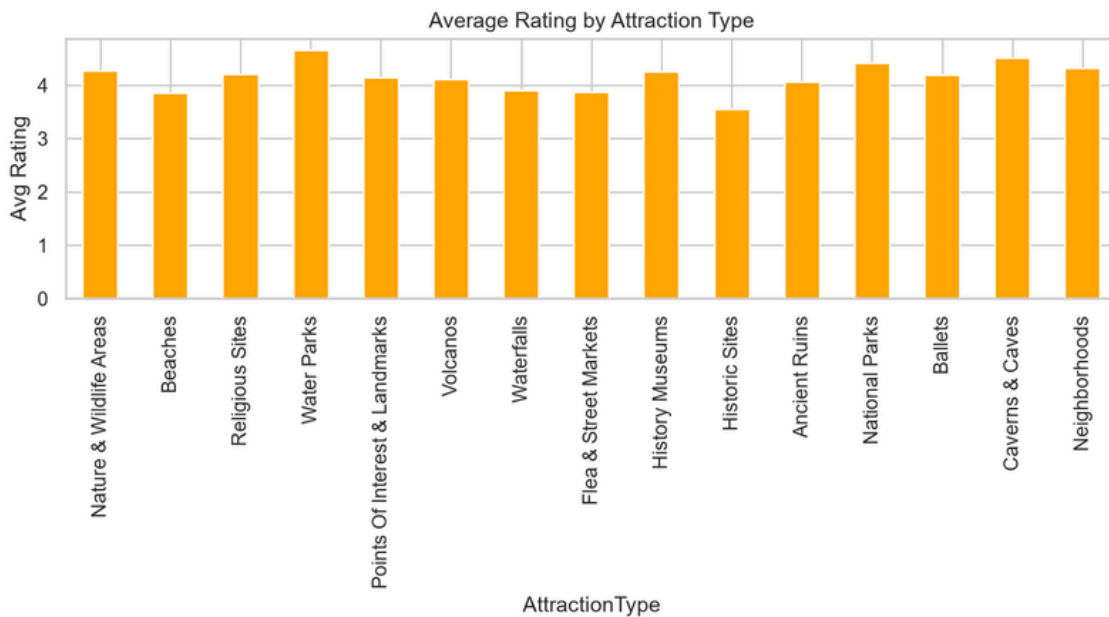
Asia leads in visit volume (about 15,600), closely followed by Australia and Oceania (about 14,800) and Europe (about 13,200). The Americas trail at roughly 8,100, and Africa accounts for under 1,000 visits, a small fraction of the total dataset.

Chart 2: Top 10 Countries by Visits



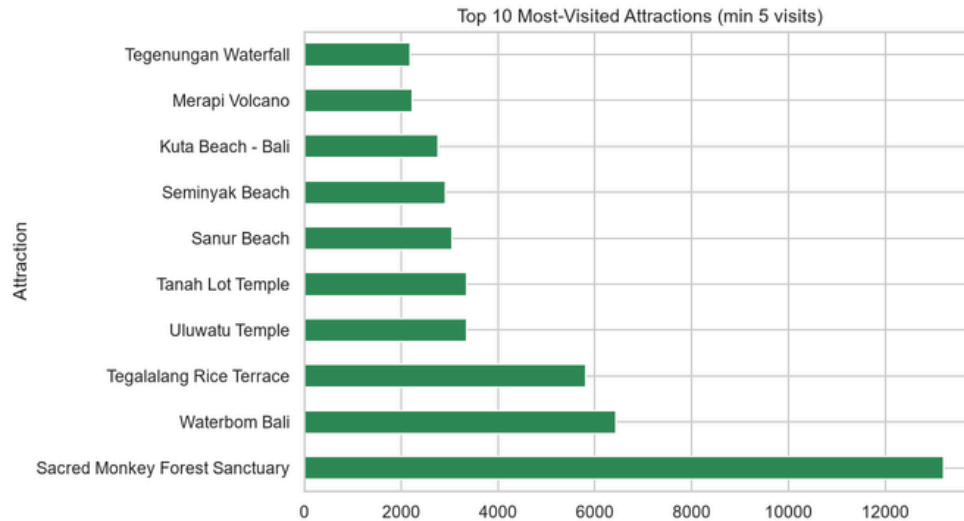
Australia alone accounts for roughly 13,000 visits, nearly double the next closest country (the United Kingdom, about 6,700). This shows the user base is heavily concentrated in a small number of source countries rather than spread evenly worldwide.

Chart 3: Average Rating by Attraction Type



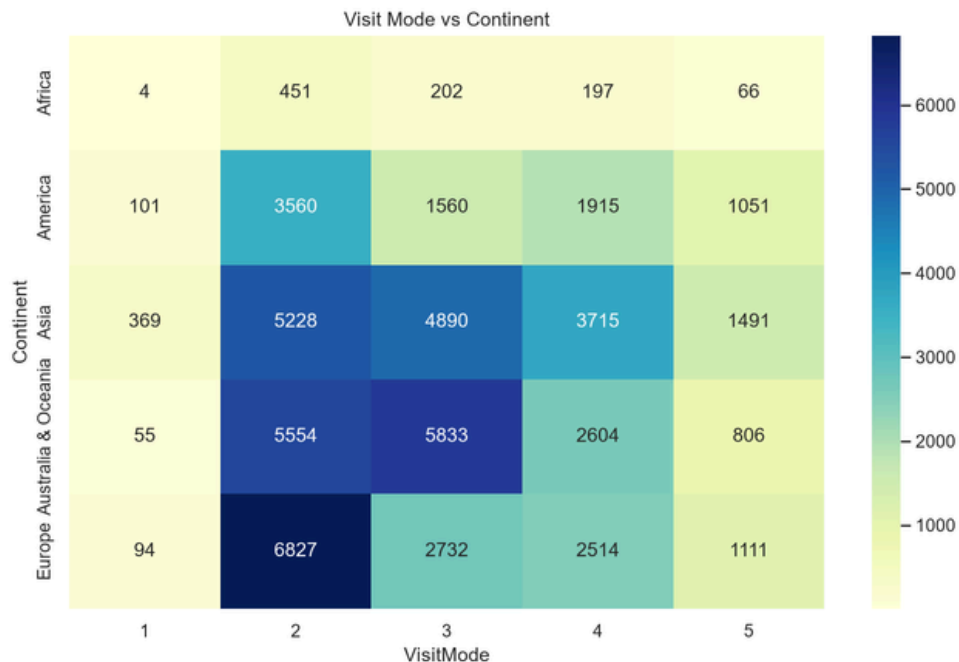
Water Parks earn the highest average rating (4.65) despite moderate traffic, while Beaches, the single most-visited category (10,917 visits), score lowest among high-traffic types (3.85). This gap suggests a service-quality opportunity: beaches may be underdelivering relative to visitor expectations.

Chart 4: Top 10 Most-Visited Attractions



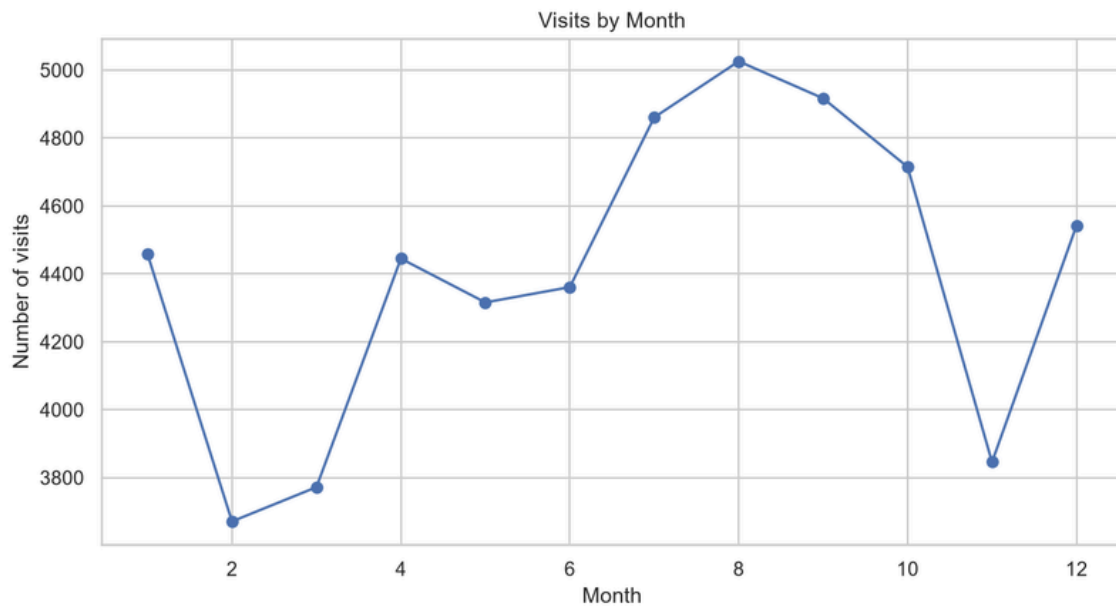
The Sacred Monkey Forest Sanctuary is the single most-visited attraction by a wide margin (over 13,000 visits), and nine of the ten most-visited attractions are located in Bali, Indonesia. This indicates the dataset's attraction inventory is heavily concentrated in one destination, even though the country-level visit counts show a broader user base.

Chart 5: Visit Mode vs Continent



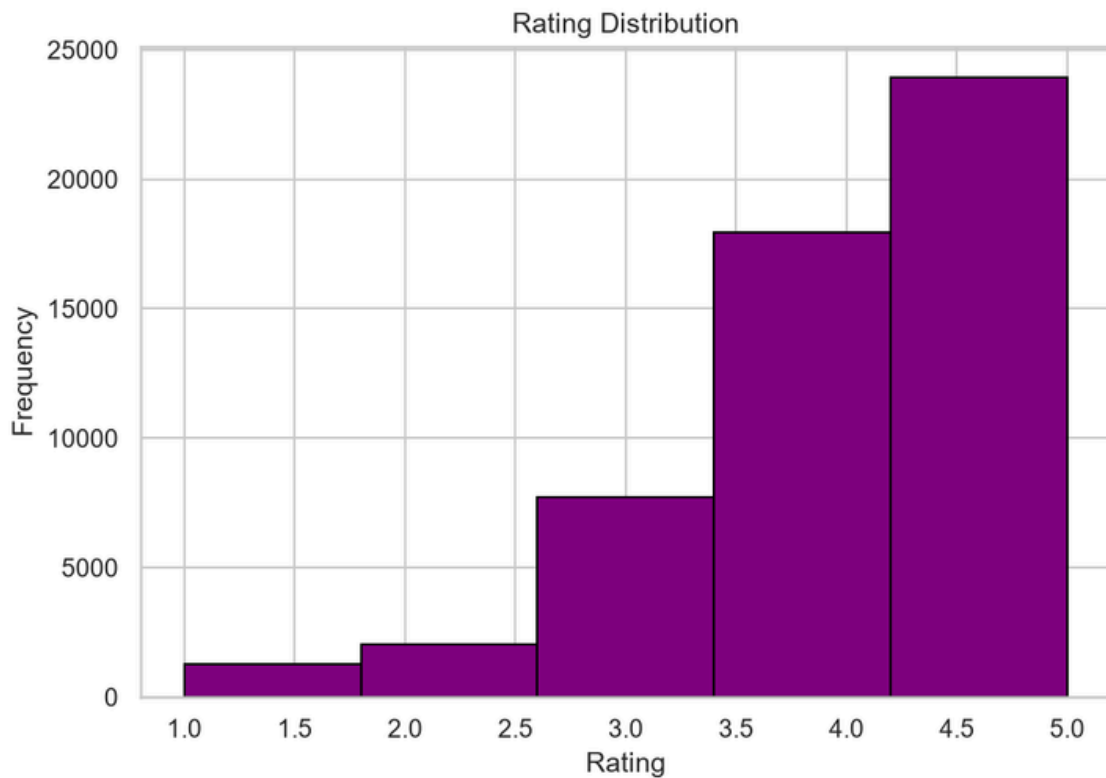
Visit mode 2 is the most common mode on every continent, ranging from 451 visits in Africa to 6,827 in Europe, while visit mode 1 is the rarest everywhere. This suggests one visit mode dominates travel behavior broadly across regions rather than visit mode varying strongly by continent.

Chart 6: Seasonality of Visits



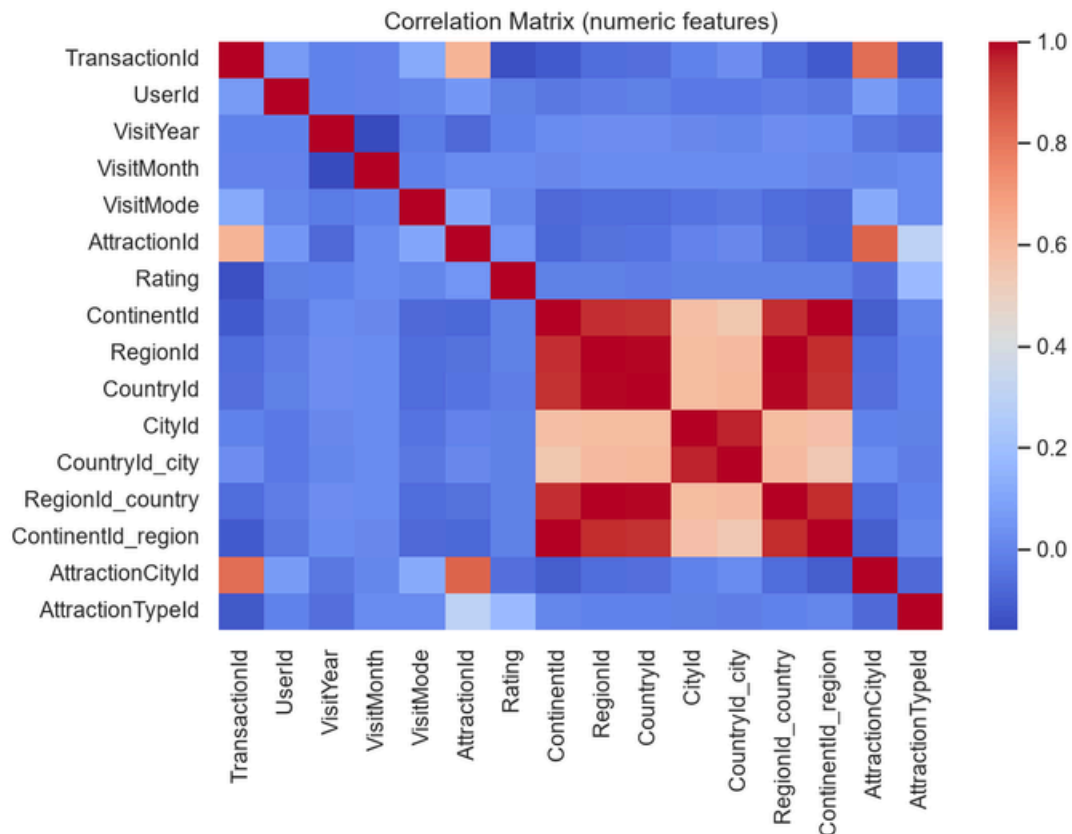
Visits peak in the middle of the year, rising sharply from June and reaching a high point in August (just over 5,000 visits), with a secondary dip in February (the lowest point, around 3,670 visits) and a smaller dip in November.

Chart 7: Rating Distribution



Ratings are strongly skewed toward the high end: the 4.2 to 5.0 range accounts for roughly 24,000 of the 52,930 ratings, and very few ratings fall below 2.5. Overall, users rate attractions positively far more often than negatively.

Chart 8: Correlation Matrix (numeric features)



Rating shows almost no linear correlation with any other numeric field, including the geographic identifiers, which instead correlate strongly with each other as expected from their hierarchical structure (Continent within Region within Country within City). This is consistent with, and helps explain, the low R^2 obtained in the regression model below.

3. Model Performance

3.1 Regression: Predicting Rating

Features used: Continent, Region, Country, City, Attraction Type, Visit Year, Visit Month.

<i>Model</i>	<i>R²</i>	<i>MAE</i>	<i>RMSE</i>
Linear Regression	0.029	0.749	0.956
Random Forest	-0.015	0.748	0.978
XGBoost (best)	0.103	0.718	0.919

XGBoost performed best but explains only about 10% of rating variance. This indicates ratings are driven substantially by factors not captured by geography or visit timing alone, most likely individual subjective experience, service quality on the day, or attraction-specific factors not present in this feature set. Adding user-level historical behavior (for example, a user's average past rating) would likely improve this in a future iteration.

3.2 Classification: Predicting Visit Mode

<i>Model</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1</i>
Logistic Regression	0.282	0.400	0.282	0.316
Random Forest (best F1)	0.440	0.463	0.440	0.448
XGBoost	0.492	0.483	0.492	0.441

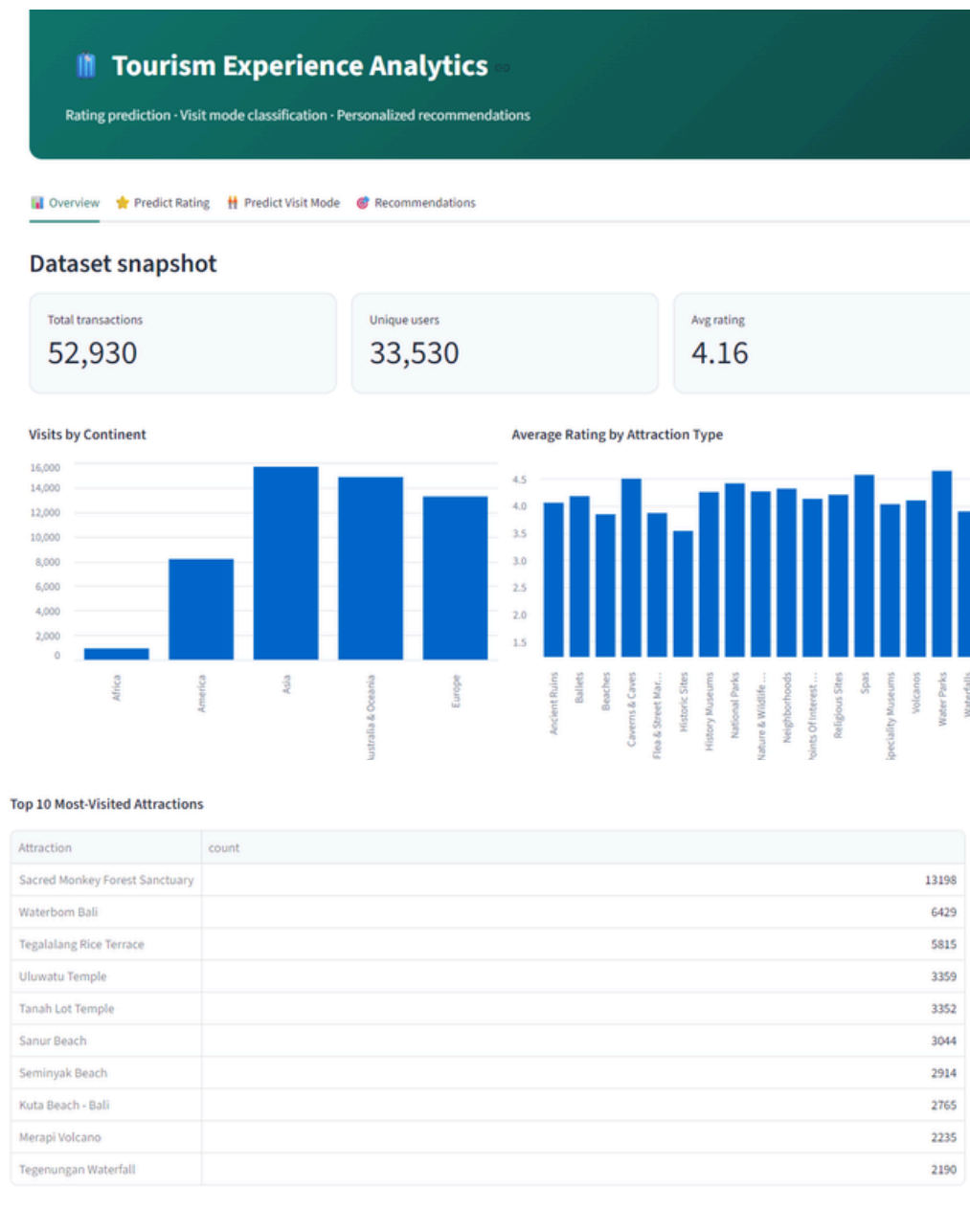
Random Forest was selected as the best model by weighted F1-score, balancing performance across the imbalanced visit-mode classes better than Logistic Regression. The minority class (Class 1, n=125) is the hardest to predict, as expected given limited training examples. As with the regression task, accuracy could likely be improved by adding user-level features such as a user's historical visit mode pattern.

6. Streamlit Application

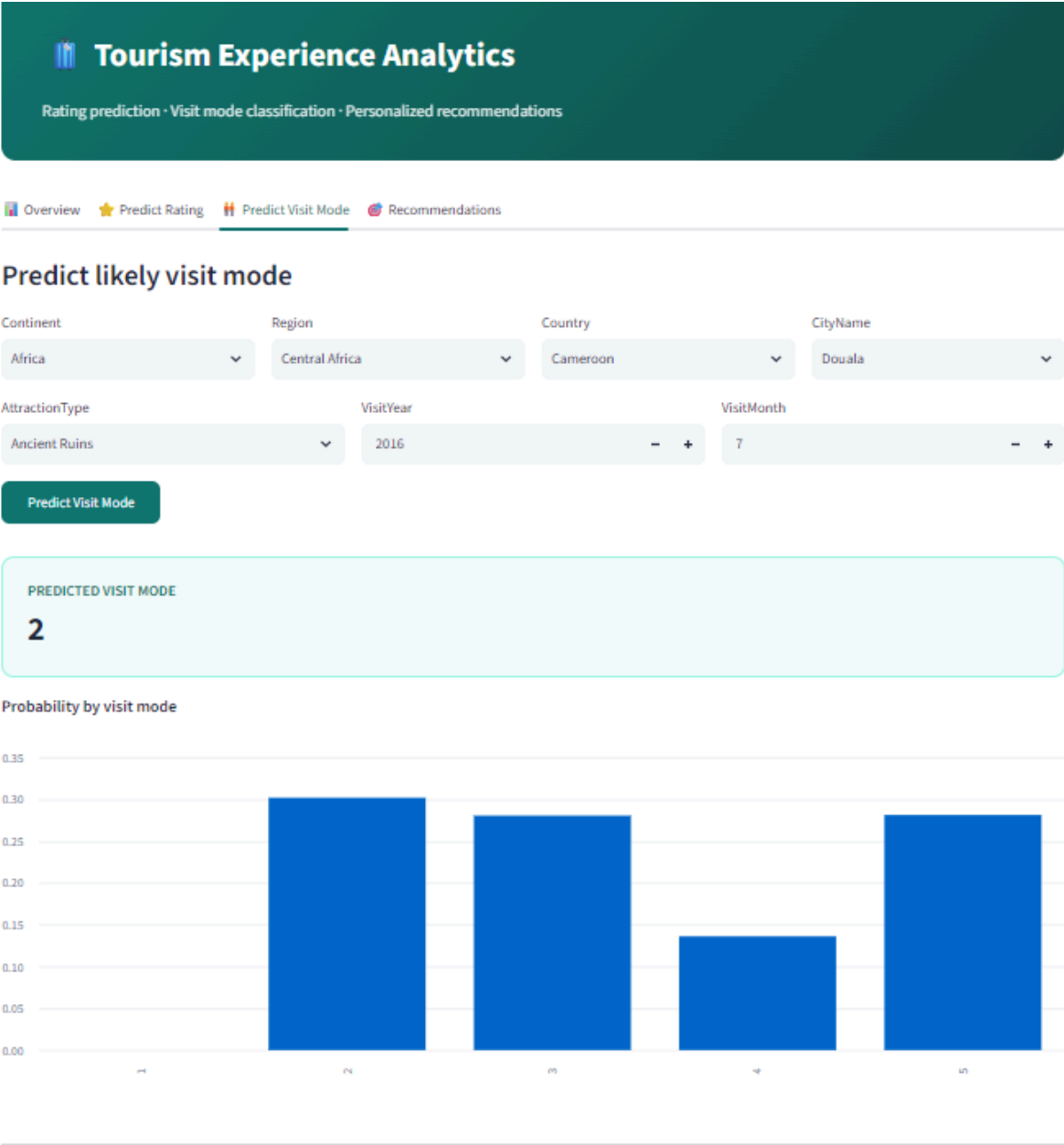
An interactive Streamlit application was built with four tabs: Overview (dataset metrics and charts), Rating Prediction, Visit Mode Prediction, and Recommendations (both collaborative and content-based). Location inputs use cascading dropdowns (Continent, then Region, then Country, then City) so users only see valid combinations at each step.

GIT HUB link: <https://github.com/Azrocks/tourism-experience-analytics>

Screenshot: Overview tab



Screenshot: Recommendations tab



Tourism Experience Analytics - Regression, Classification & Recommendation System

Screenshot: Rating Prediction result

Tourism Experience Analytics

Rating prediction · Visit mode classification · Personalized recommendations

Overview

Predict Rating

Predict Visit Mode

Recommendations

Personalized attraction recommendations

Method

☒ Collaborative Filtering (existing user)

☐ Content Based (similar to an attraction)

Select User ID

14

Number of recommendations

5

Get Recommendations

Attraction	14
Coban Rondo Waterfall	0.0031
Goa Cina Beach	0.0022
Sempu Island	0.0018
Kalibiru National Park	0.0018
Balekambang Beach	0.0016

Overview

Predict Rating

Predict Visit Mode

Recommendations

Predict the rating a user might give

Continent

Africa

Region

Central Africa

Country

Cameroon

CityName

Douala

AttractionType

Ancient Ruins

VisitYear

2016

VisitMonth

7

Predict Rating

PREDICTED RATING

4.14 / 5

7. Conclusion

This project delivers a complete, end-to-end tourism analytics pipeline: cleaned and merged data across nine source tables, exploratory analysis surfacing concrete service-quality and demand insights, two supervised models (regression and classification) with transparent performance evaluation, a working hybrid recommendation system, and a deployed interactive application. The most valuable finding may be a negative one, that geography and timing alone are weak predictors of satisfaction and behavior, which itself directs future work toward richer, user-history-based features.